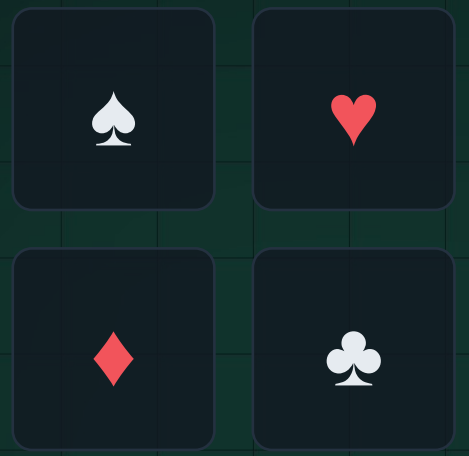
MID-TERM REVIEW · TECHNION · 2026

Learning to Play Belote with Reinforcement Learning

PPO self-play on a two-player Belote environment



Noa Curiel & Yoni Benabou

Why Belote?

A popular strategic game

One of the most played card games in France, with a Belote tournament held every year at the Technion.

A real AI challenge

Imperfect information: the opponent's hand is hidden, so every decision is a bet on what they hold and on how the trick will unfold.

Our motivation

As long-time players, we wanted to see whether an RL agent can rediscover human conventions — or beat them — purely through self-play.

Our two-player Belote variant

RULES

- 32-card deck, 8 cards per player — 16 stay hidden all round
- Trump suit drawn at random each round
- Must follow suit; otherwise must trump; must over-trump when trump is led
- Highest card wins the trick (trump beats every other suit)
- +10 for the last trick (dix de der), +20 for King + Queen of trump (belote-rebelote)

NON-TRUMP



11 pts 10 pts 4 pts 3 pts 2 pts 0 pts

TRUMP



20 pts 14 pts 11 pts 10 pts 4 pts 3 pts 0 pts

In trump, the Jack and the Nine jump to the top.

From four-player Coinche to two-player Belote

INITIAL IDEA

Four-player Coinche

- Two teams of two
- Bidding & contracts
- Full original rules

→

THE PROBLEM

Too noisy for PPO

- 3 hidden hands + a partner
- Reward shared by the team
- Bidding adds huge variance

→

OUR CHOICE

Two-player Belote

- One opponent, zero-sum
- Keeps trumps, cutting, hidden info
- Single shared network in self-play

Development pipeline

01
Rule engine
Legal moves, scoring

02
Gym env
598-dim obs, action mask

03
Baselines
4 reference agents

04
Actor-critic
Shared policy + value

05
Self-play + GAE
Experience & advantages

06
PPO update
Clipped objective

07
Evaluation
Duplicate deals

KEY DESIGN PRINCIPLE

Every stage is validated by its own test suite (7 in total) before the next one is built on it. Two silent bugs still slipped through — found by code review and by analysing why the agent did not learn, then locked in by new tests.

The Belote environment

OBSERVATION

598-dim vector

- My hand (32) + cards played (32)
- Trump suit (4), both scores (2)
- Who won each trick: me / opponent
- Per-trick detail: my card, theirs, leader

ACTION

1 of 32 cards

- Illegal cards masked to −∞ before the softmax
- The agent only ever samples a legal move

REWARD

Points actually won

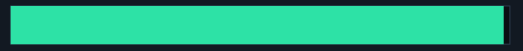
- Trick points, normalized
- + dix de der, + belote-rebelote
- No ±1 win bonus (removed — see results)

BUG FIXED

Trick winners were stored as an absolute seat (0/1) while everything else was relative to the active player — breaking the perspective a shared self-play network relies on. Re-encoded as me / opponent / not played.

Hand-crafted reference opponents

Strength = average point margin per game against the Random agent (duplicate deals, 10 000 games each)

Random	Plays a uniformly random legal card. Sanity floor.		+0.4
Heuristic	Leads its strongest trump, wins tricks as cheaply as possible, discards low.		+3.9
VoidExploit	Tracks suits the opponent can't follow and leads them to drain its trumps.		+7.3
TrumpCount	Counts remaining trumps and leads trump only when holding the majority.		+7.9

One shared network, trained by PPO self-play

ACTOR-CRITIC NETWORK

Observation	598
Dense + ReLU	128
Dense + ReLU	128

Policy	Value
32 logits	1 scalar

Same weights play both seats — observations are always seen from the active player.

TRAINING LOOP

1. **Rollout**
   128 self-play rounds, alternating who leads
2. **Credit**
   Trick reward sent back to the move that earned it
3. **GAE**
   Advantages computed per seat (γ = λ = 1)
4. **PPO update**
   Clip ε 0.2 · 4 epochs · batch 256 · Adam 3e-4

Repeat → evaluation every 50 000 games

Measuring skill in a game of luck

56%

Win rate of our best hand-crafted agent against a random player.

The deal decides most games. A plain evaluation's error bar (±3–7 pp) was larger than the progress we wanted to detect.

DUPLICATE DEALS — AS IN BRIDGE

1 deal

hands A, B
+ trump

→ **Game 1** Agent: A Opp: B

→ **Game 2** Agent: B Opp: A

Same cards, seats swapped: the luck of the deal cancels out.

PRECISION, SAME 4 000 GAMES

Heuristic vs Random	±1.44 →	**±0.74 pts**
TrumpCount vs Heuristic	±1.22 →	**±0.52 pts**
VoidExploit vs Heuristic	±1.20 →	**±0.41 pts**

4–9× fewer games for the same precision

What unlocked learning

THREE CHANGES

01 **GAE per seat**

Returns mixed both players' points — the signal "who wins the points" vanished. Bug fixed.

02 **No ±1 win bonus**

Winning a round mostly depends on the deal: the bonus was noise.

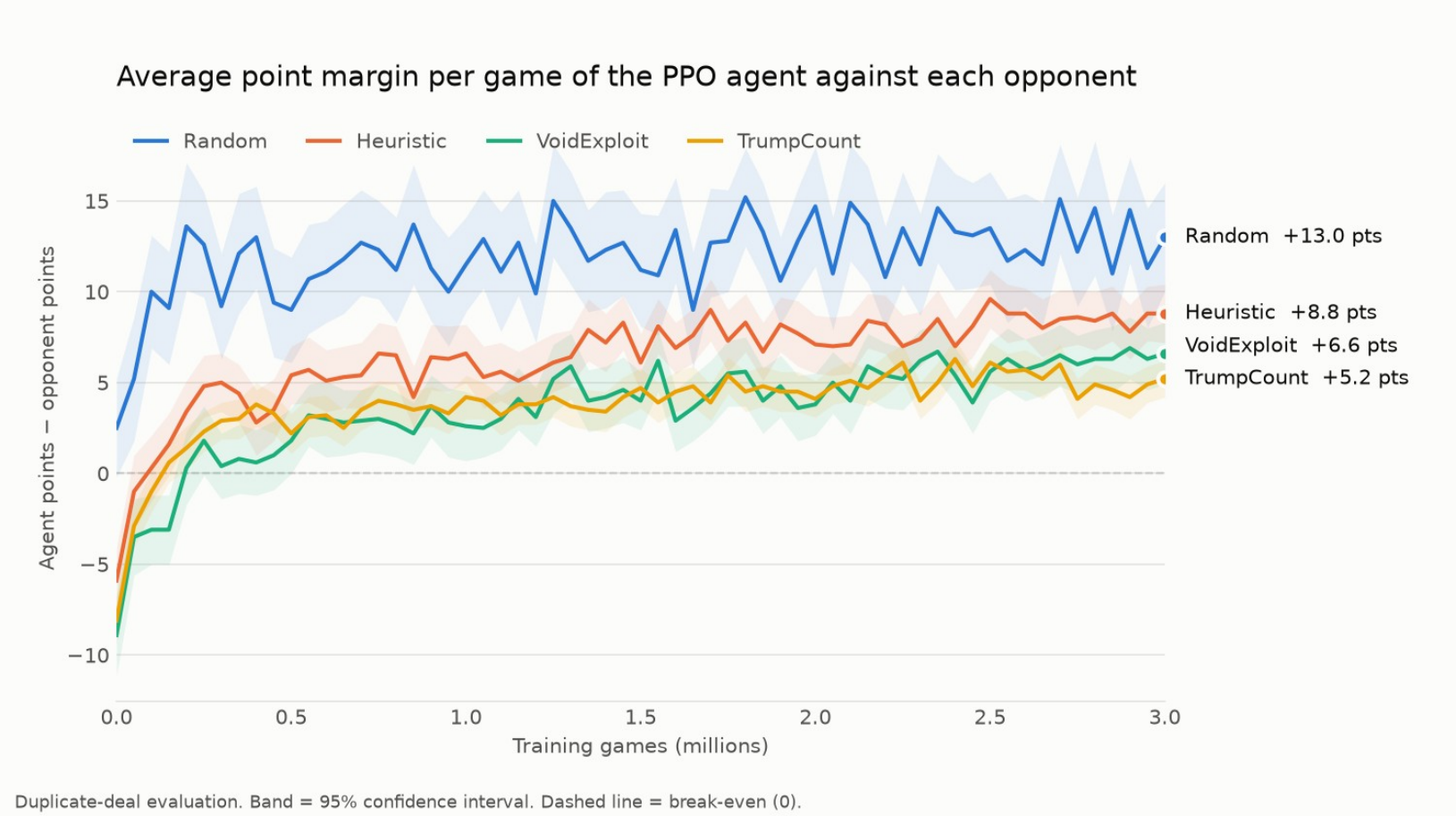
03 **Learning rate 3e-4**

1e-3 made updates too abrupt (KL ≈ 0.04).

POINT MARGIN VS TRUMPCOUNT · 200K GAMES



3 million games: the agent beats every baseline



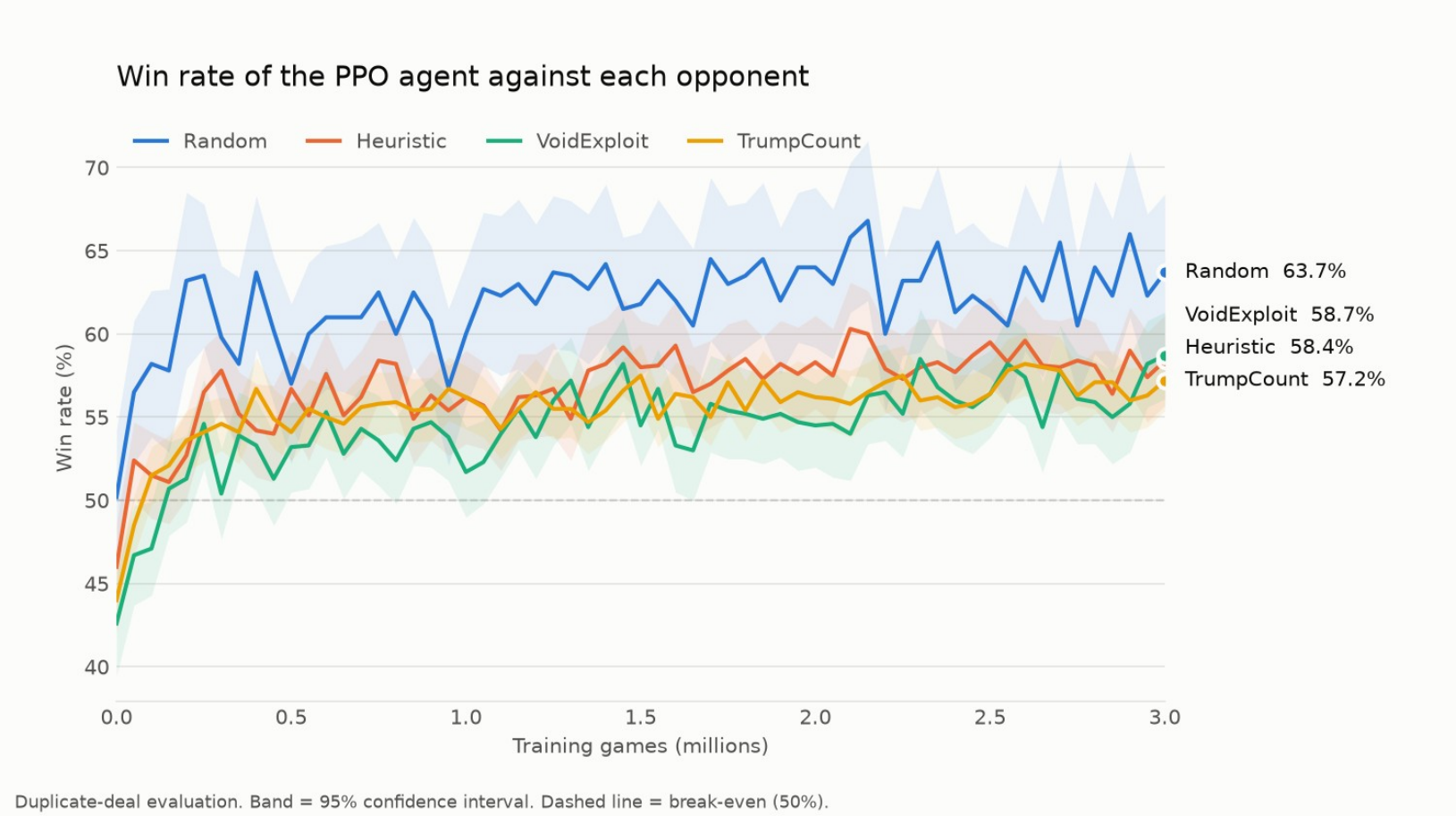
+5.0 vs TrumpCount
±0.4 · 56.0% wins

+5.7 vs VoidExploit
±0.5 · 56.2% wins

+8.3 vs Heuristic
±0.5 · 59.2% wins

+12.5 vs Random
±0.6 · 63.1% wins

Above 56% against every baseline



56.0%

wins vs TrumpCount, our strongest heuristic (±0.7 pp, final agent re-tested on 6 000 games)

WHY NOT 80%?

The deal decides most games: even TrumpCount only wins 56% against a random player.

Putting 56% in context

MATCH-UP	WIN RATE	POINTS / GAME
HAND-CRAFTED AGENTS		
Random vs Random	50.1%	+0.4
Heuristic vs Random	52.1%	+3.9
VoidExploit vs Random	56.0%	+7.3
TrumpCount vs Random	56.0%	+7.9
TrumpCount vs Heuristic	54.6%	+3.8
OUR PPO AGENT (3M GAMES)		
PPO vs Random	63.1%	+12.5
PPO vs Heuristic	59.2%	+8.3
PPO vs VoidExploit	56.2%	+5.7
PPO vs TrumpCount	**56.0%**	**+5.0**

READING THE TABLE

Random vs Random = 50%: the evaluation is fair.

Our best heuristic beats a random player by only 56%: the deal decides most games.

PPO wins 56% against TrumpCount — as often as TrumpCount wins against a random player.

Duplicate deals · ≥ 6 000 games per match-up · win rate ±0.6–0.9 pp

We kept losing to it — so why only 63%?

OUR REASONING

01 It felt much stronger

Playing against it ourselves, we lost most games: 63% of rounds won vs Random looked far too low.

02 We built a cheater

It sees the opponent's hand and plays optimally — yet wins only 76% of rounds vs Random.

03 Luck dominates one round

Even perfect information cannot rescue a bad hand.

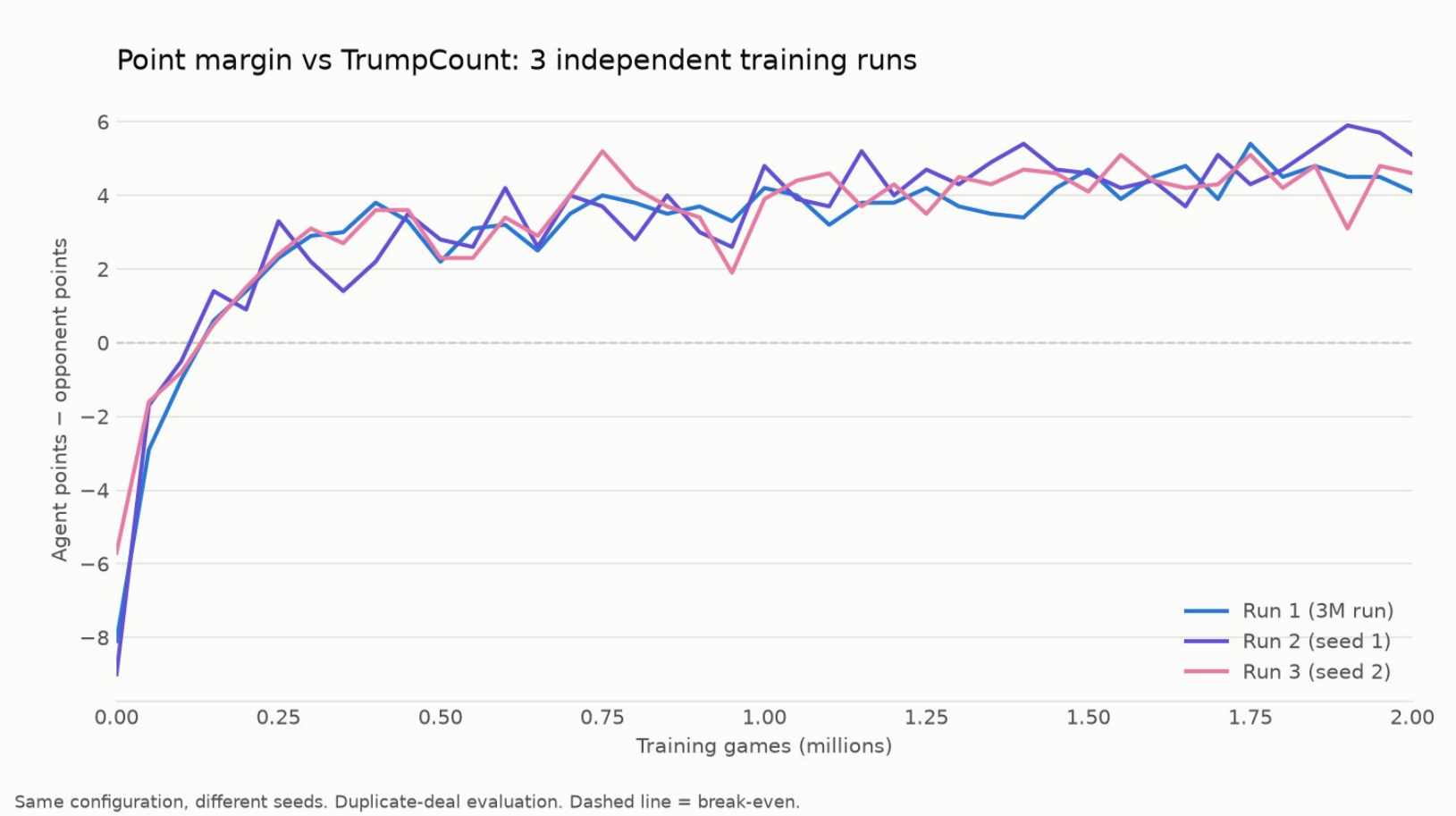
04 Evaluate like real Belote

Full games, first to 501 points (~10 rounds): luck averages out.

MATCH-UP	ROUNDS WON	GAMES WON (501)
Random vs Random	50%	51%
Heuristic vs Random	52%	62%
TrumpCount vs Random	56%	72%
PPO vs Random	63%	**83%**
PPO vs Heuristic	59%	**72%**
PPO vs VoidExploit	56%	**64%**
PPO vs TrumpCount	56%	**62%**
Cheater vs Random*	*76%*	***99%***

* sees the opponent's hand and plays optimally
Duplicate games · 1 000 per match-up (400 for the cheater) · ±2 pp

Three independent runs, the same result



VS TRUMPCOUNT AT 2M GAMES

Run 1	**+4.35**	55.9%
Run 2	**+4.94**	56.8%
Run 3	**+5.06**	55.9%

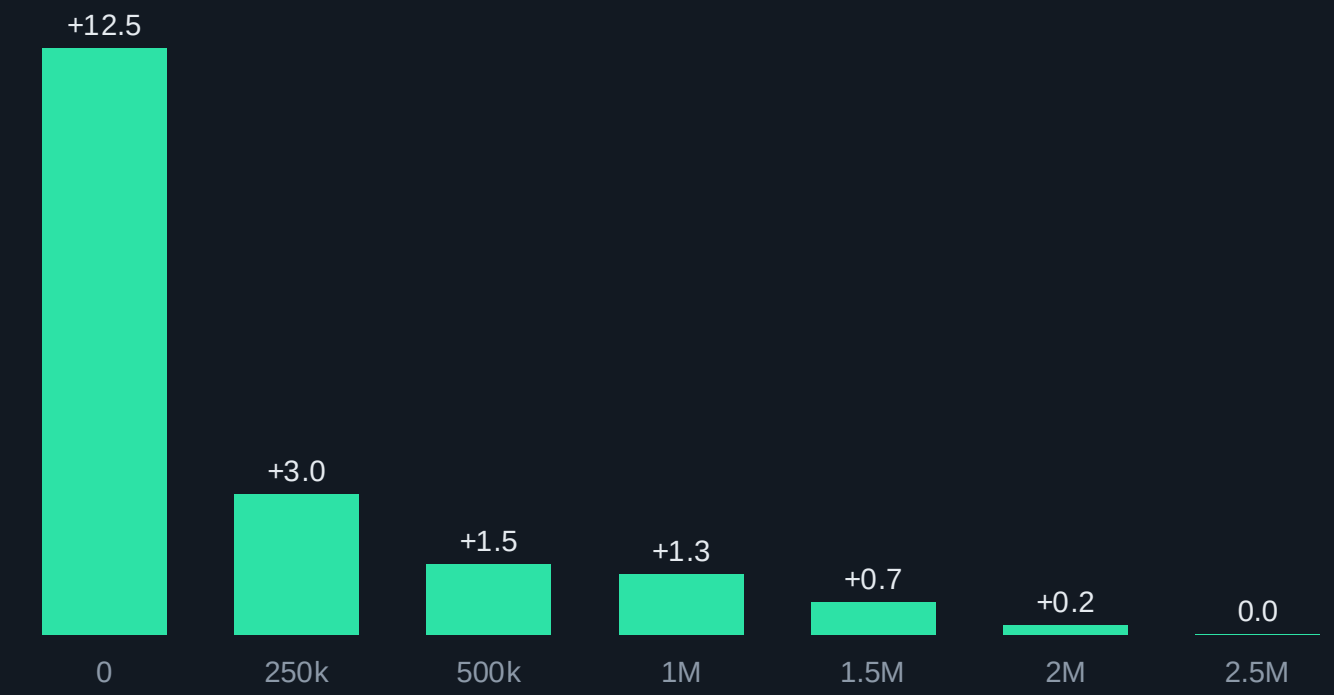
pts / game · win rate · ±0.4 pts each

+4.8 ±0.4

mean across seeds — the spread is no larger than measurement noise

Real progress in self-play — until ~2M games

FINAL AGENT VS ITS OWN EARLIER VERSIONS



Opponent checkpoint (training games) · 1 000 duplicate deals each · ±0.5–0.9 pts

TAKEAWAYS

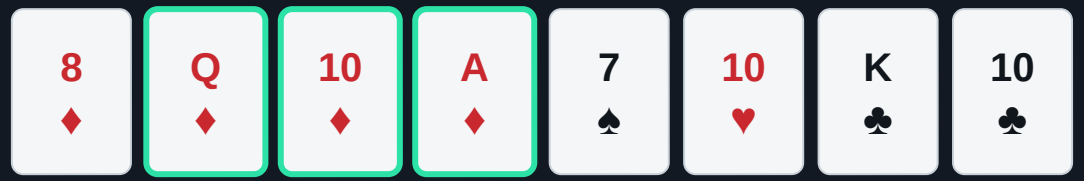
It improves against itself, not only against our heuristics.

Gains shrink, then stop near 1.5–2M games: the last million added nothing.

Going further needs a better agent (memory, bigger net, league self-play), not longer training.

What the agent learned on its own

SAME HAND, TWO PLAYERS · TRUMP ♦



Big trumps — but not the Jack or the Nine, the opponent holds both.

TrumpCount	Leads the Ace of trump at once → captured by the Jack (−31 pts).	7 pts
PPO agent	Ducks under the Jack with the 8, flushes out the Nine with the Queen, then cashes 10 + Ace as masters.	**58** pts

LEARNED ALONE

- Wait for the opponent's top trumps to fall
- Profitable ruffs
- Cash master cards at the end

STILL WRONG

Leads a Ten while the Ace is still out (−21 pts), with 100% confidence.
→ weak memory of which high cards remain.

Where we stand — and what comes next

DONE

- Tested Belote environment, 4 baselines, PPO from scratch
- Duplicate-deal evaluation: 2–3× tighter error bars
- Fixed 2 silent bugs (perspective leak, GAE across seats)
- Ablations: removing the win bonus is the key change
- 3M games: +5.0 pts / 56% wins vs our best heuristic — reproduced in 3 independent runs

NEXT

01 Memory

LSTM or a belief table of the opponent's cards

02 League self-play

Train against a pool of past versions

03 Trump choice

Let the agent bid the trump suit (Phase 4)

04 Human vs agent

Plug the trained agent into the playable CLI

QUESTIONS

Thank you

Noa Curiel & Yoni Benabou

